Image signal theory

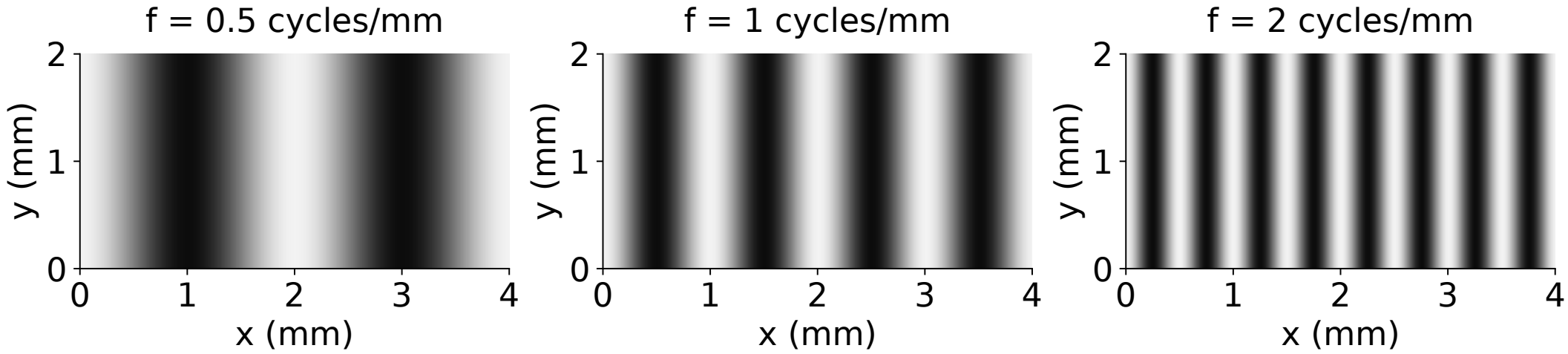
Spatial signals

An image can be represented by a function $I(x, y)$, where I is intensity.

A line through the image gives a one-dimensional intensity profile, $I(x, y_0)$. (line cut)

We apply the same sinusoidal and Fourier descriptions used for time signals, with position replacing time.

Sinusoidal resolution target



One bright-dark pair is a cycle. Increasing *spatial* frequency brings the stripes closer together.

Mean, amplitude, and phase

Sinusoidal image:

$$I(x) = \overline{I} + A\cos(2\pi u x + \phi).$$

Symbol	Meaning	Units
$\overline{I}$	Mean intensity	Normalized intensity
A	Sinusoidal amplitude	Normalized intensity
u	Spatial frequency	Cycles/mm
ϕ	Phase	Radians

The notebook uses $\overline{I} = 0.5$ and $A = 0.5C_0$, where the contrast depends on $A/\overline{I}$.

Spatial period

The period is the distance needed for one complete cycle:

$$\Lambda_x = \frac{1}{u}.$$

For $u = 2$ cycles/mm, $\Lambda_x = 0.50$ mm.

Check: a pattern has three bright-dark pairs across 2 mm. What are its spatial frequency and period?

Spatial period and frequency conventions

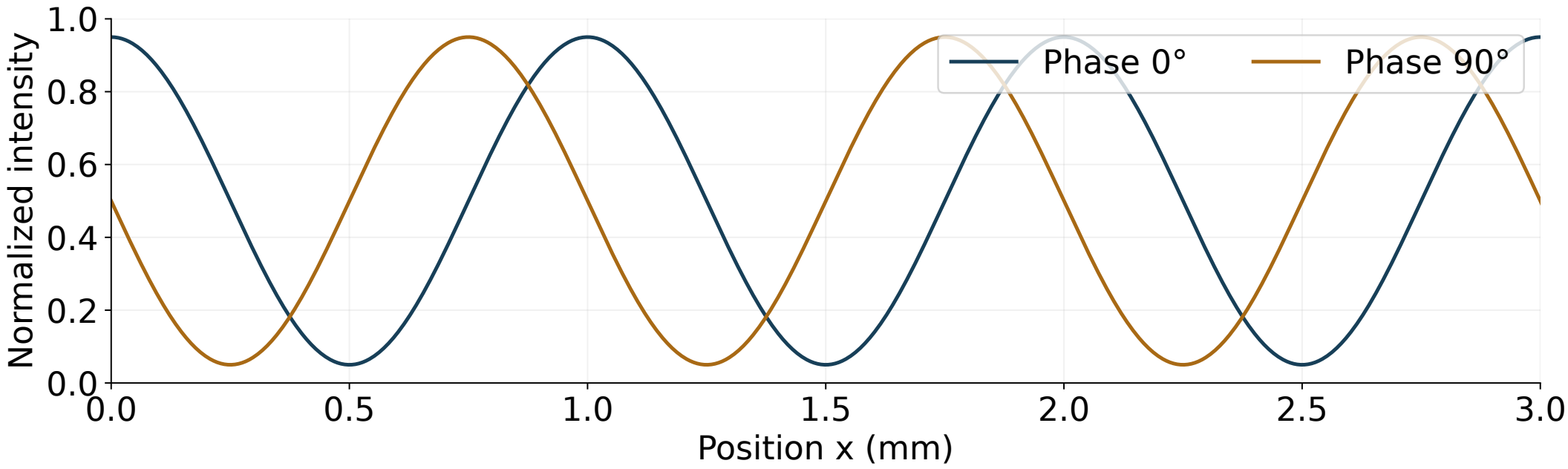
Relations between wave number (k) and frequency u.

$$I(x) = \overline{I} + A\cos(2\pi u_0 x + \phi) = \overline{I} + A\cos(k_{x0} x + \phi).$$

$$u_0 = \frac{1}{\Lambda_x} \quad \text{(cycles/mm)}, \qquad k_{x0} = 2\pi u_0 = \frac{2\pi}{\Lambda_x} \quad \text{(rad/mm)}.$$

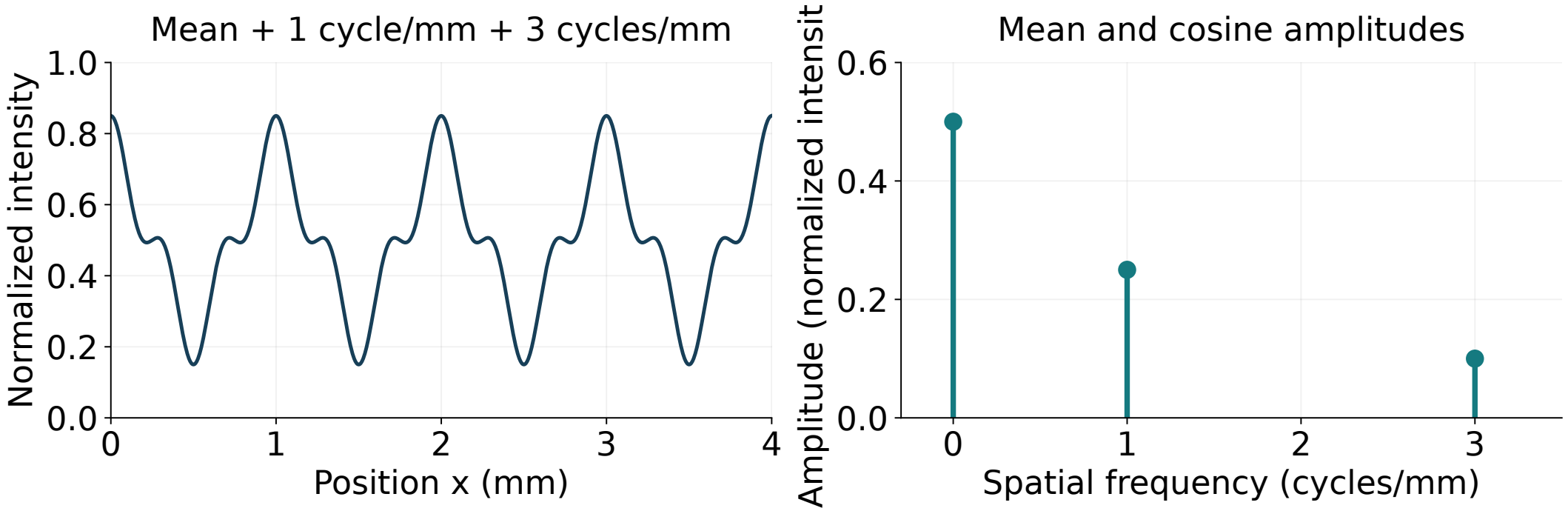
x is a position; Λ_x is a repeating distance. The notebook's frequency f is a cycles-per-length frequency, like u_0.

Phase: ϕ



Both signals have the same frequency and amplitude.

A signal as a sum of spatial frequencies



The mean, or zero-frequency component, sets the overall intensity level.

Spatial Fourier transform

For a one-dimensional profile, we use

$$\tilde{I}(u) = \int_{-\infty}^{\infty} I(x) e^{-j2\pi ux}\, dx.$$

- x is position in mm and u is spatial frequency in cycles/mm.
- Magnitude describes the strength of each frequency component.
- Phase describes its spatial alignment.

The inverse transform uses $e^{+j2\pi fx}$.

Two-dimensional Fourier coordinates

Positions (x, y) correspond to spatial frequencies (u, v):

$$\tilde{I}(u, v) = \iint I(x, y) e^{-j2\pi(ux+vy)}\, dx\, dy.$$

With angular spatial frequencies,

$$\hat{I}(k_x, k_y) = \iint I(x, y) e^{-j(k_x x + k_y y)}\, dx\, dy.$$

$$k_x = 2\pi u, \qquad k_y = 2\pi v.$$

If x, y are in mm, then u, v are in cycles/mm and k_x, k_y are in rad/mm.

Magnitude and phase of a complex spectrum

Write the Fourier transform as a real (R) and imaginary (I) component.
$F(u,v) = F_R(u,v) + jF_I(u,v)$, where $j^2 = -1$.

$$|F| = \sqrt{F_R^2 + F_I^2}, \qquad \angle F = \text{atan2}(F_I, F_R).$$

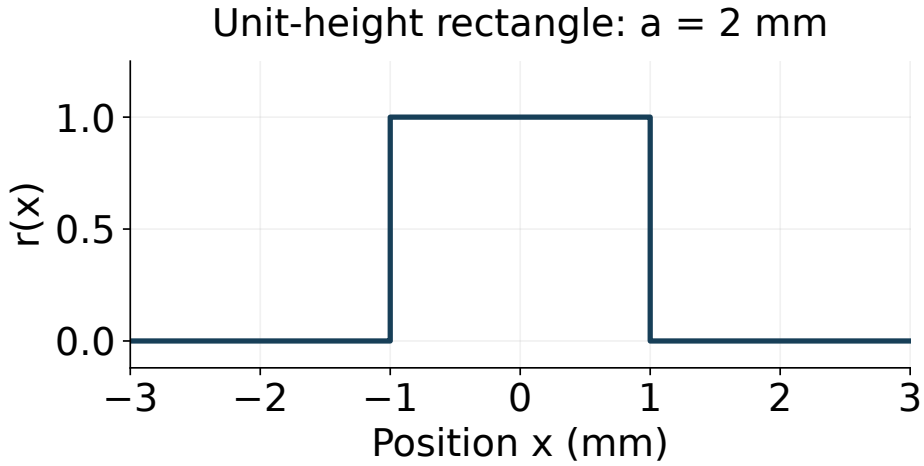
The two-argument angle function identifies the correct quadrant. Phase is undefined where $|F| = 0$.

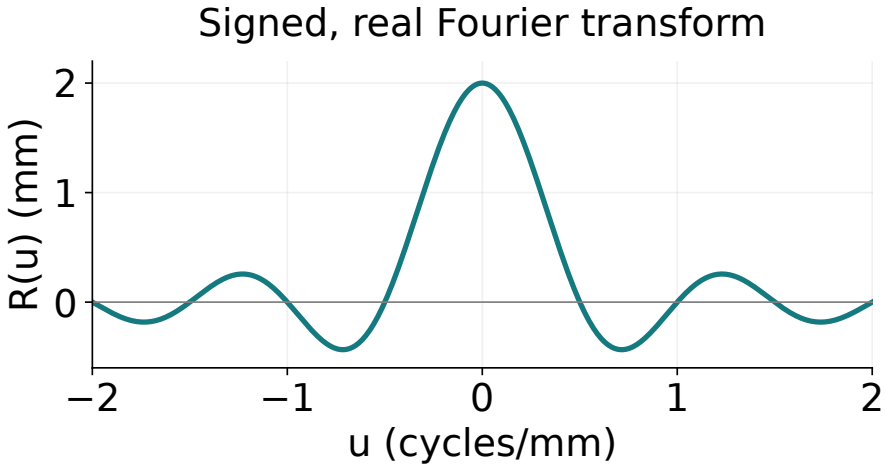
Example: for $F = -1 + j\sqrt{3}$, the magnitude is 2 and the phase is $2\pi/3$ radians, or $120°$.

A rectangular spatial signal and its transform

A unit-height rectangle $r(x) = \text{rect}(x/a)$ equals 1 for $|x| < a/2$ and 0 outside. Its width is $a > 0$. A pixel's sensitivity across its width is often modeled as a rect function.

$$R(u) = a\ \text{sinc}(au), \qquad \text{sinc}(z) = \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}(0) = 1.$$

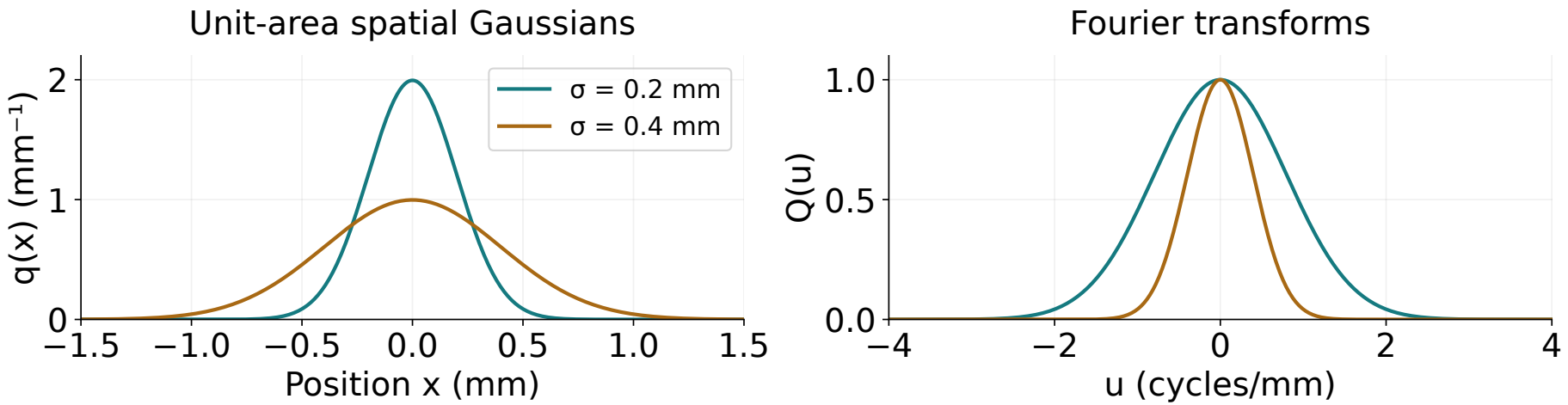




A Gaussian spatial signal and its transform

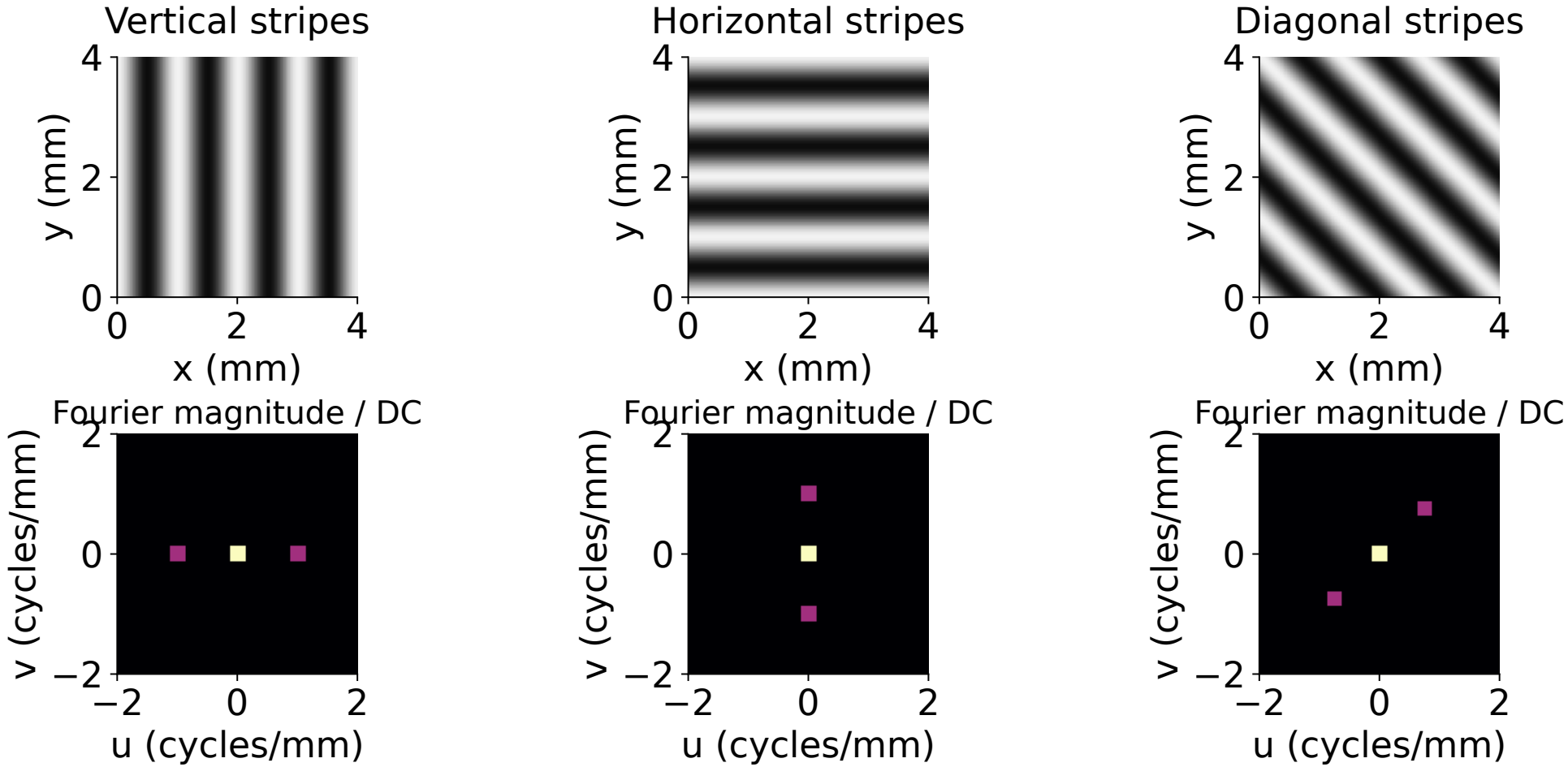
For a centered Gaussian with unit area,

$$q(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)}, \qquad Q(u) = e^{-2\pi^2\sigma^2u^2}.$$



A wider spatial Gaussian has a narrower frequency response. The Gaussian is a common approximation for a camera's point-spread-function (PSF). A greater σ implies more blur.

Frequency has a direction in two dimensions



The frequency vector points in the direction of intensity variation, perpendicular to the stripes.

Spatial sampling

For ideal point samples with spacing p,

$$I_n = I(np), \qquad u_s = \frac{1}{p}.$$

n is an integer sample index (i.e., pixel number). p is the pixel pitch - the center-to-center spacing between neighboring pixels.

The sampling frequency u_s is the number of samples per mm.

For a sinusoid, the number of samples per period is

$$N_{\text{period}} = \frac{\Lambda_x}{p} = \frac{1}{pu}.$$

Nyquist frequency

$$f_N = \frac{f_s}{2} = \frac{1}{2p}.$$

For an ideal band-limited signal, recovery from equally spaced samples requires its highest frequency to be strictly below f_N.

A sinusoid with a frequency below this limit is sampled more than two times per period.

Example: $p = 0.25$ mm gives $f_s = 4$ samples/mm and $f_N = 2$ cycles/mm.

A sampling limit in angular-frequency units

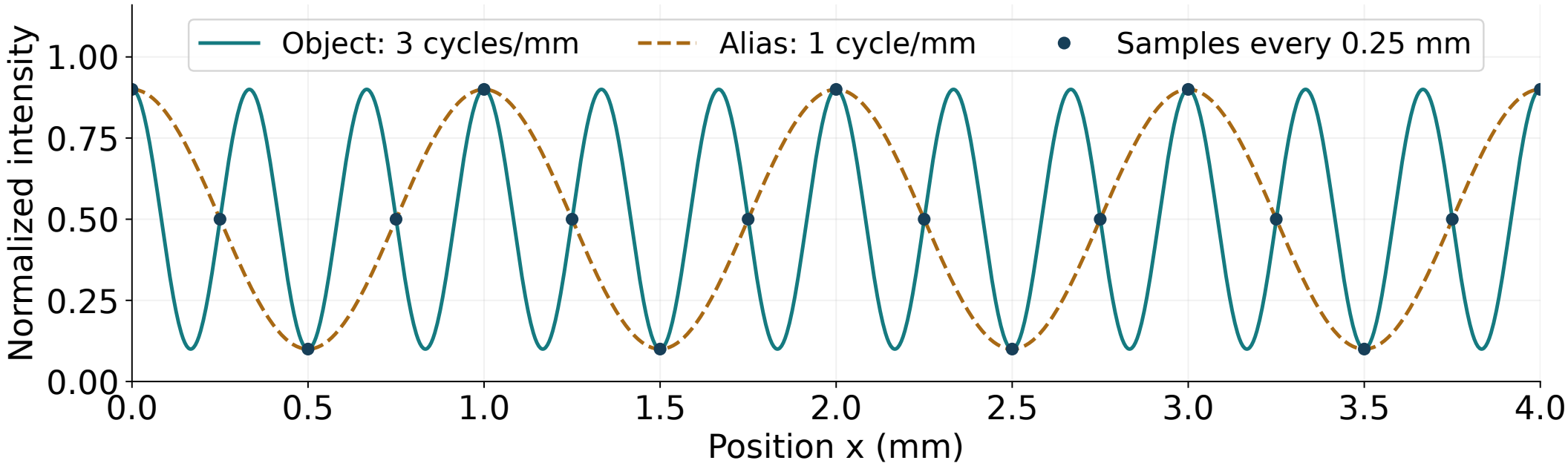
For pixel spacing p, the angular sampling frequency is $k_s = 2\pi/p$ and the angular Nyquist frequency is $k_N = \pi/p$.

$$k_{\text{max}} < \frac{\pi}{p}, \qquad p < \frac{\pi}{k_{\text{max}}}.$$

Example: if $k_{\text{max}} = 10\pi$ rad/mm, then $u_{\text{max}} = 5$ cycles/mm and the limiting pitch is 0.10 mm.

Choose a smaller pitch for phase-independent sampling of the highest-frequency component. Exactly at the limit there are only two samples per period.

Aliasing



At these sample locations, the two continuous signals give identical values. This is aliasing.

A sampling calculation

A detector records a 1.5 cycles/mm stripe pattern.

Quantity	$p = 0.20$ mm	$p = 0.40$ mm
Sampling frequency f_s	?	?
Nyquist frequency f_N	?	?
Samples per stripe period	?	?
Target below Nyquist?	?	?

Calculate the entries and predict whether either detector demonstrates incorrect stripe spacing due to aliasing.

Sampling and physical blur

An object can contain frequencies above a detector's Nyquist frequency.

Physical blur reduces high-frequency contrast before sampling. This can limit aliasing, but also reduces fine detail.

Increasing the displayed image size adds no new detector measurements.

Next class: the point-spread function (PSF) and modulation transfer function (MTF) quantify these effects.

Discussion questions

1. What changes when spatial frequency doubles: stripe period, amplitude, or both?
2. What is the Nyquist frequency for $p = 0.10$ mm?
3. Why can two different continuous patterns agree at every recorded sample?

Reading and resources

Prince and Links, *Medical Imaging: Signals and Systems*: spatial signals, Fourier representations, and sampling.

- MIT RES.6-007: lecture notes on convolution, Fourier transforms, sampling, and interpolation.
- MIT HST.582J: lecture notes for further biomedical signal examples.